ZEROTH COHOMOLOGY IS A HOMOLOGICAL FUNCTOR ON THE HOMOTOPY CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3843-3850, immediately following the statement at 3834-3841.

Editorial note. The reduction is the span's own step; the ses exactness is the terminal input.

Allowed standard prerequisites (editorial).

- Long exact cohomology sequence for termwise split short exact sequences. Homology, Lemma homology-lemma-long-exact-sequence-cochain, cited at proof lines 3849-3850. Terminal input: exactness of H^0(A) -> H^0(B) -> H^0(C) for a termwise split short exact sequence of complexes.

Source reference. Original reference 0117: focused-seeds/geometry/S9-der-cohomology-homological/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. The functor*

$$H^0 : K(\mathcal{A}) \longrightarrow \mathcal{A}$$

is homological.

Proof. Because H^0 is a functor, and by our definition of distinguished triangles it suffices to prove that given a termwise split short exact sequence of complexes $0 \to A^\bullet \to B^\bullet \to C^\bullet \to 0$ the sequence $H^0(A^\bullet) \to H^0(B^\bullet) \to H^0(C^\bullet)$ is exact. This follows from Homology, Lemma 0117. □

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